A 3D PRINTED HYBRID SAXOPHONE MOUTHPIECE FOR DATA COLLECTION

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ABSTRACT

In the past few decades there have been explorations on hybridization of acoustic and electronic instruments in order to achieve new musical possibilities. These kind of hybrid instruments, also known as hyperinstruments, rely on the use of new technologies, sensors, cameras, microcontrollers, and other devices, as well as computers and software, to collect and process data to generate new audio. This paper documents the authors' own exploration on hyperinstruments and the search to access new instrumental techniques for the saxophone, and offers a detailed description of the design of a 3D printed saxophone mouthpiece which combines acoustic and electronic characteristics to collect data from the performance, aiming to control live electronics. The mouthpiece's performance is characterised and the potential to use the collected data for pedagogical purposes is also considered.

1. INTRODUCTION

In recent years, 3D printing technology has allowed new developments in design throughout a wide spectrum of fields. In the field of music and acoustics, this new technology has also opened new possibilities allowing for accessories, instruments, resonators, replacement parts, protective cases and other related items to be developed. Perhaps the importance of 3D printing technologies for musical research lies on the fact that these new printers can produce prototypes and final products with a relative ease of construction, unlike working in the old-fashioned way by handcrafting the instruments or parts out of natural materials (typically wood). It is well known that luthiers, woodwind and brass instruments makers, as well as other professionals in similar fields, learn their craft over many years, and a musician or researcher is usually not skilled to build musical instruments, repair them, build new parts or developments, therefore slowing down any projects in this field. Not only are the new 3D printers allowing more professionals to access easy and fast prototypes, but their low price has opened the doors for the DIY amateur enthusiast.

There are documented projects of printed models that have furthered the developments of musical instruments and acoustic investigation. The authors catalogue these 3D printed models in six categories: protective accessories (shells, cases, pads, caps, etc), repairments (knobs, string pegs, bridges and other small part replacements), enhancements (pickup holders, shoulder rests, inserts, etc.), replicas, customizations and innovative designs. Many of these models, mostly from DIY projects, are shared as creative commons in specialized webpages such as thingiverse.com, yeggi.com, grabcad.com and many others. Other scholarly projects focus on researching musical and acoustic issues rather than practical ones, such as producing microtonal flutes (Dabin et al. 2016) or a multi-sized shank in a trumpet mouthpiece (Leitman and Granzow 2017) for acoustic inquiry.

This current project takes advantage of the 3D printing technologies with a model that fits in the category of customizations, aiming to collect data on the air pressure blown into the saxophone. This mouthpiece, along with other developments, will be used in the creation of a hypersaxophone with the goal of allowing saxophonists to access new instrumental techniques.

This paper examines previous developments by other authors, and the influence those projects had on the development of the mouthpiece. Detailed information on the design and tests is also discussed to offer an overview of the project and its potential, and possible future uses of the mouthpiece.

2. RELATED WORK

In recent decades, new enhancements have been added to musical instruments, generating two new categories of musical instruments: hybrid instruments and hyperinstruments. Hybrid instruments are those instruments that combine different materials in their construction, characteristics of two or more instruments, or different ways of producing sound (e.g. electroacoustic instruments). On the other hand, in hyperinstruments the audio output is processed by a logical system loaded in a computer or microprocessor. There have been many examples in which hyperinstruments or hybrid instruments have been developed by means of combining traditional acoustic instruments and new technological developments such as sensors, microprocessors and computers.

In previous works, researchers have demonstrated that it is possible to collect data on different aspects of instrumental performance through the use carefully designed systems. As Matthew Burtner described it, these

systems "convert the performance data into a continuous data stream" (Burtner 2002) which could be used to control digital audio processing, to study the performative practices, to understand the acoustic phenomena or to gather information useful for pedagogical purposes. Despite the fact that there have been a good number of projects dealing with woodwind or brass instruments, very few of these designs deal with the issue of air flow through the body of the instruments and how to quantify the energy carried by the air.

Gathering data from air flow can be very difficult as disturbing the air column inside a musical instrument results in undesired affectations to the overall acoustics of the instrument. In consequence, a design where all the data collection happens away from the air column running inside the instrument is the only feasible option. This, however, raises different problems on different aerophones as the way the air is blown into the instrument varies. In instruments with a blowhole, such as flutes, the air is blown across a sharp edge of the blowhole. This system allows for external data collection of the airflow. In whistles, single reed instruments, double reed instruments, brass instruments (cup mouthpiece) and pipes, acquiring this information can be difficult as the air travels directly from the mouth to the inside of the instrument.[1]

In research undertaken at McGill University, it was proven how difficult but useful it can be to gather air flow information and reuse it as a musical control variable. In this 2005 project, air pressure sensors are mounted around the mouthpiece gather the data. However, this system can be unreliable since the flute performative practices sometimes require a change of the angle in which the air is blown into the instrument, resulting in variation of the air pressure blown into the sensor (Da Silva, Wanderley, and Scavone 2005).

Other projects aiming to measure the energy of the air stream utilize a microphone which gathers information on the quality and dynamics of noise produced by air to obtain comprehensible data of the energy carried by the air stream. The use of microphones to obtain information on the air blown has been very successful in projects such as Smule, an iOS app which has been successfully used by thousands of users around the globe (Wang 2009).

Projects such as the SABRe (Schiesser and Schacher 2012) deal with problems related to reed instruments. In this case, it becomes impossible to use a microphone or an air pressure sensor inside the body of the instrument or at the end of it since foreign objects inside the instrument would disturb the air column, and placing the data collector at the end of the instrument does not yield accurate results since the air stream's energy is lost through the various holes of the instruments depending on the fingerings. In this particular project, its authors developed a system in which a small hose mounted on top of the instrument redirects a portion of the air stream into an air pressure sensor, obtaining accurate readings in every situation. In this case the sensor, being mounted on top of the mouthpiece, is close enough to the air stream source (player's mouth) to ensure that no energy (pressure) is lost before the sensor reading is done.

Another way to achieve accurate acquisition of data was explored in a joint effort between Universitat Pompeu Fabra and Standford University, where it was demonstrated that modifications to the mouthpiece of an alto recorder to redirect a portion of the air flow into an air pressure sensor was possible (Garcia et al. 2011) by creating conduits in the body of the mouthpiece to redirect a portion of the air stream into a sensor.

Based on these previous developments the authors have modified a 3D model of an alto saxophone mouthpiece (shared on thingiverse.com) to produce a new version that features modifications that allow a portion the air stream to be redirected into an air pressure sensor, thus obtaining accurate readings without disturbing the air column inside the instrument at the same time as it provides a comfortable playing experience.

3. SYSTEM OVERVIEW

The hybrid saxophone mouthpiece has been designed with the purpose of offering a tool that can be used both to perform and to collect data about the performance. The collected data can eventually be used in different situations which may include information gathering for performance study or more practical goals such as controlling diverse parameters inside a software capable of communicating via serial or MIDI protocols.

3.1. 3D Design

The first step in developing the mouthpiece was considering the fact that introducing an object inside a saxophone causes disturbance in the air column, therefore changing the quality of the sound, affecting the intonation, producing "squeaks", or completely stopping the sound. Based on past projects, listed above, the authors decided to work with a 3D design that would feature an alternate conduit in the body of the mouthpiece that would permit data collection without affecting the air stream flowing through the mouthpiece and final air column inside the saxophone.

The mouthpiece is an adaptation of the *Alto Saxophone Mouthpiece* designed by Thingiverse user Allanarps (Allanrps 2011). The original design is based upon a Yamaha beginner mouthpiece, with similarities including bore length and shape, facing width, tip opening and basic length measurements, featuring a flat baffle and squared throat (bore).

The adaptation keeps the basic shape of the interior with the same dimensions in the chamber of the mouthpiece, facing width and bore shape, which ensures that the air column running through the chamber behaves in the same manner as in the original design. The changes made to the design include a shorter tip opening of 1.8mm

[1] For a more detailed classification of aerophones: (von von Hornbostel, Sachs, and Montagu 2011)

(original is 2mm) and a thicker beak that allocates a small conduit running through the body of the mouthpiece to an added section in the back in which a sensor is mounted to measure the air pressure. See Figure 1 for a comparison between the original design by Allanarps and the adaptations made to it.

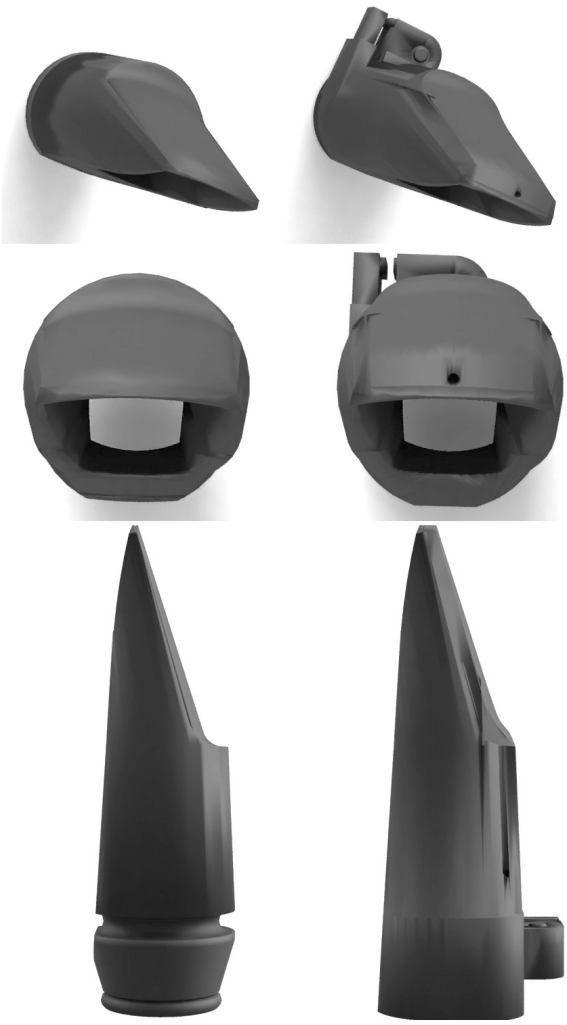


Figure 1. Comparison between the original 3D model (left) and the new version (right) with adaptations to allocate a pressure sensor and an alternate air conduit.

As mentioned before, the conduit inside the body was designed to allow comfortable playing with no disturbance to the air stream and air column, and to achieve this goal a minimal amount of material was added to the top of the beak of the mouthpiece to accommodate the conduit. In total, 2mm of thickness was added to the beak, of which a 1mm radius cylindrical opening is found at the front/center of the beak, where the structure is weaker because of its dimensions. Further back, the conduit opens more to fit the dimensions of the sensor at its back end (see Figure 2). There is also a small side canal that allows moisture and saliva to escape the system. This opening can be blocked or opened if desired with wax or tape. Further testing will determine whether this side opening is necessary.

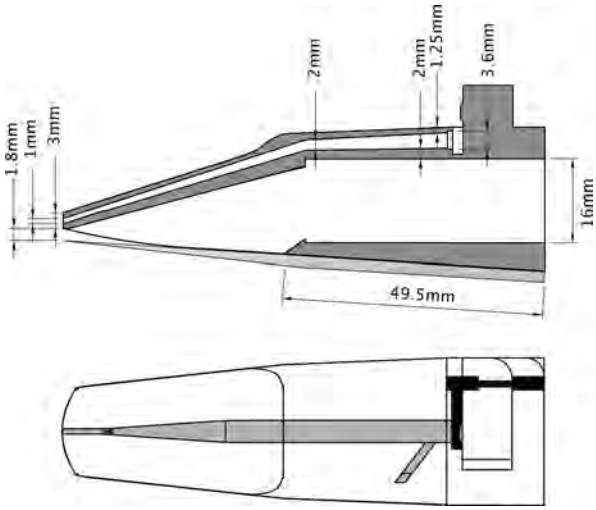


Figure 2. Interior of the mouthpiece. The top image shows a slice of the mouthpiece where the chamber and conduit are shown in white. The bottom image shows the conduit and side canal in gray. The area designed to position the sensor breakout is colored in black.

3.2. Components

The Hybrid Saxophone consists of two main components: the 3D printed mouthpiece and an electronic barometer sensor BMP180.

The 3D model, previously described, can be printed in a variety of materials, including organic-based (wood) and other plastic and polymer based materials, many of them resembling ebonite, a common material used in the fabrication of mouthpieces. For the first printing trials, an Up Mini filament deposition modelling 3D printer was used. For the printing material, ABS (acrylonitrile butadiene styrene) has been used due to its durability, ease of use and price.

To measure the air pressure, various kinds of sensors were considered. Unlike previous projects, this one has the advantage of 3D design customization which allows the opportunity to fit any reasonably sized sensor. The chosen sensor is a BMP180 designed by BOSCH as a barometric pressure sensor commonly used in applications such as indoor navigation, GPS-enhancement for dead-reckoning, slope detection, etc., sport devices, (e.g. altitude profile), weather forecast, vertical velocity indication (rise/sink speed) and others (“BMP180” 2017). Considering the applications for which the BMP180 was designed, this is not the most obvious choice of sensor, however, due to the flexibility of design it was easy to fit this sensor in the mouthpiece and match it with the end of the alternate air conduit.

The BMP180’s most relevant features are: 300-1100 hPa pressure range, an average current consumption of

650µA that can go down to 3µA, supply voltage VDD of 1.8V to 3.6V, a relative accuracy pressure of ±0.12 hPa, and the use of the I^2C protocol at a 3.4MHz transfer rate (BOSCH 2015).

Unlike most sensors designed to measure air pressure, the BMP180 doesn't feature a funnel-like structure to redirect the air into it, which becomes unnecessary with the conduit built into the mouthpiece 3D model (see Figure 3). The BMP180 can be purchased on its own or mounted on different breakout boards. For this project, the GY68 breakout has been chosen because on this design the "hole" that captures the air in the sensor is situated close to one of its edges, which makes it convenient to introduce it in the body of the mouthpiece and reach the end of the conduit. Other breakout boards place the sensor on the center of the PCB, and the use of these boards would translate in carving into the model to reach the air conduit, this could compromise the structure of the mouthpiece.



Figure 3. BMP180 Breakout GY68 next to a common air pressure sensor with a funnel-like casing.

The breakout stays in place thanks to a small protuberance on the mouthpiece which matches the screw hole in the breakout. Also, the breakout pins help secure the sensor in place once they are connected to standard 2.54mm connectors. In Figure 4, the sensor is mounted on the printed model, showing the way it "locks" in place. The GY68 breakout design is perfect for handling and replacing the sensor if necessary as it only needs to be pushed into the opening in the mouthpiece designed to allocate it. Therefore, any user can replace it without the assistance of a technician and at a very low cost.



Figure 4. BMP180 Breakout GY68 secured on the mouthpiece. The visible silicone on the breakout protects its components from moisture without affecting the sensor response.

3.3. Board and On-board software

At this stage of the project, the barometric sensor is working together with a Teensy LC board[1], sending data using the I^2C protocol. This board is also capable of communicating using the MIDI protocol without having to implement complex code or hardware development, which opens up the possibility of utilizing the system as a MIDI controller.

The program loaded on the board executes a self-calibration when the system is powered. To calibrate, it firstly determines the ambient temperature, and then sets a current pressure (actual atmospheric pressure) as a baseline value, setting that value to 0. Once these parameters have been established, every iteration of the program loop compares the new results to the baseline value, outputting positive numbers for any value greater than the baseline, and negative numbers to values under the atmospheric pressure. The values are expressed in hectopascals.

The sensor is affected by temperature in a manner similar to woodwind instruments. To address this, the program performs a new calibration every five minutes to compensate for temperature changes that occur due to breathing and blowing into the mouthpiece. The sample rate is at 33 readings per second, but can be changed according to the needs of the project. A faster sample rate could be beneficial to detect attacks more precisely and avoid latency in musical contexts. Another fact to be considered when deciding the sample rate is the delay

[1] For more information on the Teensy LC, visit the developer's website at https://www.pjrc.com/teensy/teensyLC.html

that occurs between the moment when the performer starts blowing and the sonic event, which depends on dynamics, expressivity, the hardness of the reed, the calibration of the instrument's mechanics (to avoid air leaks) and other factors. Further investigation must be done to address this issue.

4. TESTS AND RESULTS

As mentioned before, the goal of designing this mouthpiece is to be used as a key element of a hypersaxophone. For this reason, the performance of the mouthpiece has been tested by using it in common musical practices. In order to test the system, a Max7 patch was designed (see Figure 5). This program receives the incoming data using the USB port. The program updates the data 33 times per second in synchronization to the Teensy's firmware. Then, the data is used to create a graphic profile of the measurements in a Cartesian system representing time (in seconds) and pressure (in hectopascals). The object used in Max7 to create the visuals representing the air pressure is [plot~] set to draw a profile with 33 points per second (indicated by the markers on the X axes) thus creating a representation without smoothing the collected data. The range in both the X and Y axes can be resized to draw a profile that meets the needs of the tests.



Figure 5. Max patch designed to read incoming data.

The characterization program also features a representation of the audio input to compare the morphology of sound (attack, decay, sustain and release) with the pressure readings. It has a monitoring function as well as a recording function which allows users to save an audio file and a screenshot of the soundwave and pressure profile representations for further analysis.

In an experiment made with the help of saxophonists Reuben Chin and Jasmine Lovell-Smith, the system was tested at the Lilburn Studios at Victoria University of Wellington. The setup consisted of the mouthpiece/sensor mounted on the alto saxophone, along with a Zoom H2 microphone, both connected via USB to a Macbook Pro running Max7 on OSX 10.11. The results demonstrated the efficacy of the system using a sensor that is sensitive enough to capture minimal changes in pressure such those produced by the performer's breathing patterns. In Figure 6 one can see the rapid response of the system and the high pressure produced by preforming the slap tongue technique and a regular note in forte dynamic.

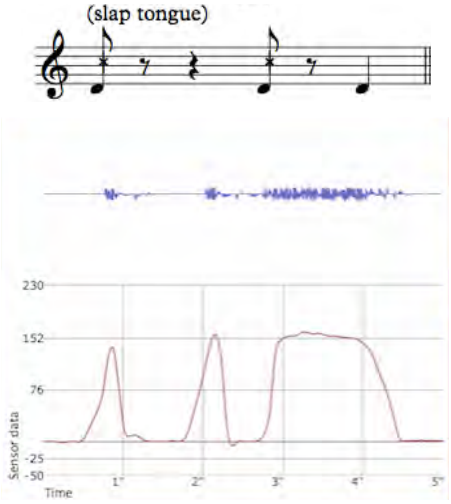


Figure 6. Slap tongue test.

In Figure 7 one can see the pressure change needed to accurately tune notes as the performer ascends or descends within an octave while playing a diatonic scale.

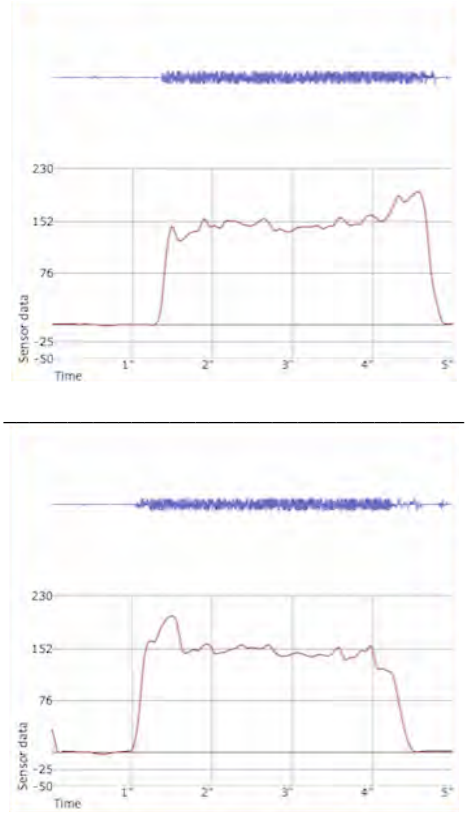


Figure 7. Ascending scale (top) and descending scale (bottom) between B4 and B5.

The system is also capable of demonstrating performance errors to be corrected, which holds promise for future pedagogical contexts. In Figure 8 the graphic shows a well-executed flutter tonguing technique and a failed execution of the same exercise. This example shows how a greater air pressure does not permit the correct execution of this technique.

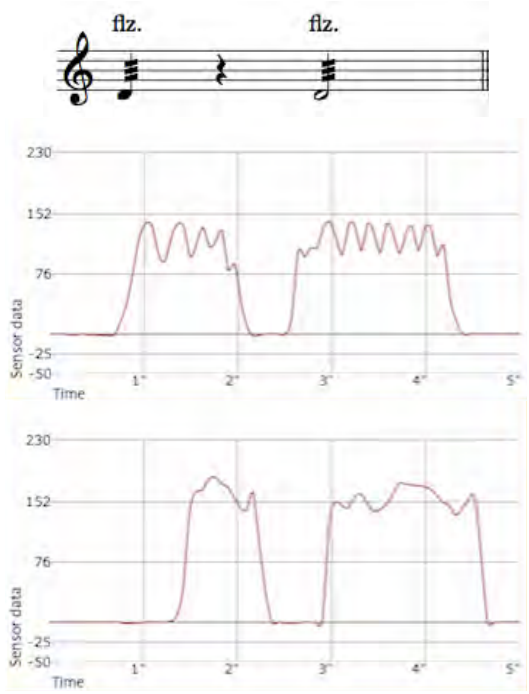


Figure 8. Fluttertonguing exercise (top) and the corresponding sensor reading of good (middle) and poor (bottom) execution of it.

The timbre of the saxophone doesn't seem to be greatly affected when the 3D printed mouthpiece is used. There are changes in the overall spectral content, however this differences in timbre are common even between two mouthpieces of the same brand and model.

The 3D printed model was compared to a Yamaha beginner mouthpiece (in which the 3D model is based). The results show variations of the energy on some of the overtones. Despite these differences, the overall overtone structure of the timbre is constant. In the lower register the differences are minimal, resulting in an almost imperceptible change of timbre. In the higher register the 3D printed model tends to present less energy on the overtones, which results in a less bright sound, but still consistent, stable and controllable. These differences are visually represented in Figure 9 and Figure 10, where a comparison of spectrum profiles is presented. These profiles where obtained from recordings of a Conn 6M (Naked lady) alto saxophone using the 3D printed model and the Yamaha mouthpiece playing a $D_{\flat 3}$ (Figure 9) and $D_{\flat 5}$ (Figure 10).

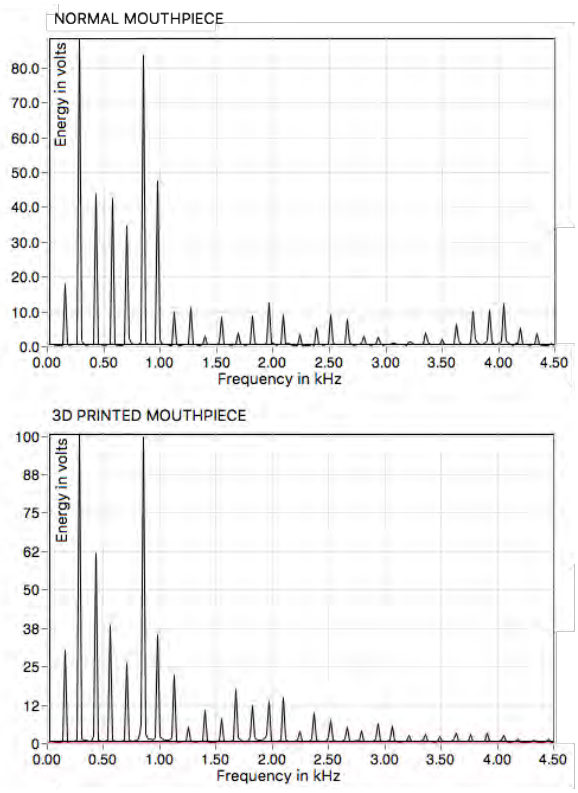


Figure 9. Overtones found in beginner Yamaha and 3D printed mouthpieces playing $D_{\flat 3}$

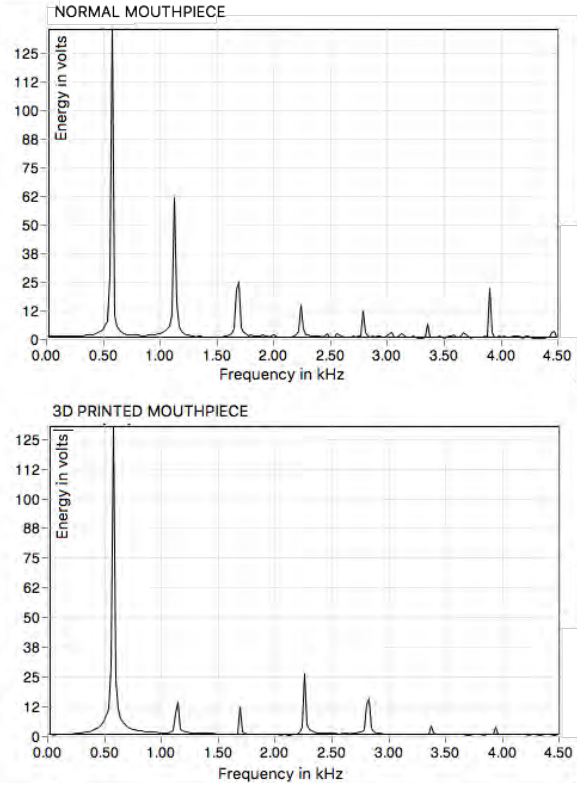


Figure 10. Overtones found in beginner Yamaha and 3D printed mouthpieces playing $D_{\flat 5}$

5. EVALUATION AND LIMITATIONS

Up to this point, the mouthpiece design together with the barometric sensor has been successful. Nevertheless, the project has faced some problems such as poor 3D printing and sensor failure due to moisture.

Achieving a good printing quality can be difficult and it does not only rely on the printer: other factors such as material and, most importantly, 3D model design matter. The resolution at which some 3D printers can print can make it very difficult to print fine details such as the 1mm radius opening of the alternate air conduit. On the first printing attempt, this section of the conduit was completely blocked. To fix this problem it was necessary to do a manual perforation. The texture achieved by the printer was somewhat rough. The major problem with this texture is the fact that it could produce air leaks in between the mouthpiece and the lips of the performer, which not only can be uncomfortable for the instrumentalists but could also make the sound production more difficult. The 3D printed model texture was removed by sanding carefully and adding a coat of bee wax. The wax not only helped smoothing the surface but also gave the mouthpiece a more organic feeling, but most importantly helps reduce any disturbance of the air flow in the interior of the mouthpiece. This coating will eventually wear off, but the mouthpiece can easily be re-coated if necessary.

The preliminary results show a consistency of playing throughout the range of the instrument with a relative ease of sound control. Even though the beak is slightly thicker than the beak on a conventional mouthpiece, according to the 3 performers surveyed, it does not seem to be uncomfortable or to pose any issues for an experienced performer. More tests are needed to assess the repercussions that using this mouthpiece has on tone quality and harmonic content. Further printing attempts will consider these results, as well as the possible differences caused by printing the model with different materials and printing techniques.

The construction of the barometric sensor permits its use despite the presence of moisture. However, the breakout GY68 fails as soon as moisture is in contact with its components as the design does not feature a protective casing. The first tests proved that, after a few minutes of playing, moisture from breathing and saliva make the system fail. To correct this issue, silicone was applied over the entire surface of the breakout and edges of the sensor, leaving the front face of the sensor uncovered to allow the air to reach the internal components of the sensor. Three possible improvements will be explored in the future. The first one will be to attempt a more precise design in which the back opening of the conduit fits perfectly with the sensor face such that no air, moisture or saliva can escape to the breakout components. This solution would be ideal because it would make replacement of the sensor an easy task of pulling out the breakout and pushing in a new one, but it can be very difficult as it depends on the quality of the printing and each new model would be slightly different. A second solution is to use a 3D printer capable of mixing two materials in a model and redesign the model in a way that the back opening of the conduit features a silicone material that can press against the sensor in such way that it prevents moisture ingress. The third solution, and possibly the easiest to achieve, is to also print a silicone casing for the breakout which could be placed on the breakout before it is set into the mouthpiece; if the sensor failed at some point it would be easily replaceable by a new sensor utilizing the same silicone casing.

6. CONCLUSIONS AND FUTURE WORK

The project has been successful so far, achieving accuracy greater than anticipated. Although it may need to be further developed, the mouthpiece is at a stage where it can be used to perform and to offer information about the performance. The system has the potential to add a new feature to control live electronics: it can not only detect the air blowing into the mouthpiece, but also is capable of detecting air suction (represented as negative values under the baseline value set on calibration).

This mouthpiece can be used in various applications. The main purpose in the development of this mouthpiece is to use it as a tool to gather data that will be used in performance practices as part of the development of a hypersaxophone. More mouthpieces based on the same idea and design will be produced to fit different instruments of the saxophone family. Changes in the design will explore the tone quality production by changing the characteristics of the baffle, bore, chamber, etc. The data collected from the pressure sensor will be used in conjunction with other sensors to control live electronics and to alter the acoustics of the instrument itself.

A new software to gather information from the data will be created to make the study of performance more effective, so that the project can be used for pedagogical purposes. It will also feature a reed with an integrated sensor to further investigate the relationship between the tongue and air pressure, and the effects that controlling them has on performance.

The use of 3D printed tools combined with the new generation of highly accurate sensors holds promise for further developments, considering the wide variety of materials that can be used to print the models. The ease of manipulation of these models offers a fast and accurate way of prototyping, developing and creating final products that will enhance our interaction with musical instruments, allowing performers and composers to expand and realize their creative ideas beyond the current limits of instrumental techniques.

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